

Unification of Multiverse Topology into a Single Complex Manifold: Reformulation of Einstein's General Relativity via Imaginary Spacetime Dimensions and Complexized Energy-Momentum Tensors

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ABSTRACT

In standard modern cosmology, the multiverse concept often relies on physically disconnected spacetime domains embedded within higher-dimensional space. In this paper, we show that parallel cosmic domains can be naturally understood as phase slices of a single four-dimensional complex spacetime manifold $M_C = M_R (+) i M_I$. By extending Einstein's metric tensor into complex space $z^\mu = x^\mu + i y^\mu$, the metric assumes a Hermitian form $g_{\mu\nu}(z) = \text{Re}(g_{\mu\nu}) + i \text{Im}(g_{\mu\nu})$. We prove that what appear to be separate parallel universes correspond simply to orthogonal phase angles θ in $[0, 2\pi)$ within one continuous complex geometry. Reformulating the field equations as $G_{\mu\nu}(z) + \Lambda g_{\mu\nu}(z) = (8\pi G / c^4) T_{\mu\nu}(z)$ resolves spacetime singularities at black hole horizons and initial cosmic states, smoothly regularizing them at the Planck scale. Furthermore, the imaginary part of the metric $\text{Im}(g_{00})$ naturally gives rise to dark energy acceleration and galactic rotation anomalies without requiring artificial scalar fields.

Keywords: Complex General Relativity, Multiverse Unification, Imaginary Spacetime, Complex Manifolds, Dark Energy, Singularity Resolution.

I. INTRODUCTION

Albert Einstein's General Theory of Relativity revolutionized our understanding of gravity by describing spacetime as a four-dimensional curved pseudo-Riemannian manifold. The field equations, written as:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G / c^4) T_{\mu\nu} \quad (\text{Eq. 1})$$

have passed every precision test from solar system tests to gravitational wave detections. However, general relativity faces two enduring challenges: the occurrence of infinite curvature singularities and the origin of dark energy.

Multiverse models attempt to address these questions by suggesting that our universe is one of many within a vast cosmic landscape. Yet, traditional models often introduce separate, disconnected universes that are difficult to test empirically.

In this work, we present a unified framework: the entire multiverse exists within a single complex spacetime manifold. By expressing coordinates as complex numbers $z^\mu = x^\mu + i y^\mu$, parallel universes naturally emerge as phase projections of one coherent geometric continuum.

II. MATHEMATICAL FRAMEWORK OF COMPLEX SPACETIME MANIFOLDS

We represent coordinates as complex quantities $z^\mu = x^\mu + i y^\mu$ in C^4 , where x^μ represents observable physical dimensions and y^μ represents internal imaginary dimensions.

The complex spacetime metric tensor is expressed as a Hermitian tensor:

$$g_{\mu\nu}(z) = \text{Re}(g_{\mu\nu}(z)) + i \text{Im}(g_{\mu\nu}(z)) \quad (\text{Eq. 2})$$

where Hermiticity requires $g_{\mu\nu} = \text{conjugate}(g_{\nu\mu})$. The complex line element is:

$$ds^2 = g_{\mu\nu}(z) dz^\mu d(\bar{z}^\nu) \quad (\text{Eq. 3})$$

The extended Christoffel symbols $\Gamma^{\lambda}_{\mu\nu}$ govern affine connections on complex manifolds:

$$\Gamma^{\lambda}_{\mu\nu}(z) = (1/2) g^{\lambda\sigma} [(d g_{\sigma\mu} / d z^{\nu}) + (d g_{\sigma\nu} / d z^{\mu}) - (d g_{\mu\nu} / d z^{\sigma})] \quad (\text{Eq. 4})$$

From these connections, the complex Riemann curvature tensor $R^{\lambda}_{\mu\nu\sigma}$ is derived as:

$$R^{\lambda}_{\mu\nu\sigma} = (d \Gamma^{\lambda}_{\mu\sigma} / d z^{\nu}) - (d \Gamma^{\lambda}_{\mu\nu} / d z^{\sigma}) + \Gamma^{\lambda}_{\kappa\mu} \Gamma^{\kappa}_{\nu\sigma} - \Gamma^{\lambda}_{\kappa\nu} \Gamma^{\kappa}_{\mu\sigma} \quad (\text{Eq. 5})$$

III. IMAGINARY EINSTEIN FIELD EQUATIONS & SINGULARITY RESOLUTION

Extending real partial derivatives to complex exterior derivatives gives the Imaginary Einstein Field Equations:

$$G_{\mu\nu}(z) + \Lambda g_{\mu\nu}(z) = (8\pi G / c^4) T_{\mu\nu}(z), \text{ where } z^{\mu} = x^{\mu} + i y^{\mu} \quad (\text{Eq. 6})$$

Smoothing Horizon Singularities:

In a standard Schwarzschild metric, the time metric component $g_{00} = -(1 - r_s/r)$ diverges as r approaches zero. Under complex coordinate extension $r \rightarrow r + i \epsilon$ (where ϵ is set to the Planck length l_P):

$$g_{00}(r + i \epsilon) = - (1 - (r_s / (r + i \epsilon))) = - (1 - (r_s r / (r^2 + \epsilon^2)) - i (r_s \epsilon / (r^2 + \epsilon^2))) \quad (\text{Eq. 7})$$

As r approaches zero, the absolute magnitude of the metric component remains strictly bounded:

$$\lim_{r \rightarrow 0} |g_{00}(i \epsilon)| = \sqrt{1 + (r_s^2 / \epsilon^2)} < \infty \quad (\text{Eq. 8})$$

This proves that complex coordinate extensions remove physical singularities, keeping curvature invariants finite everywhere.

IV. MULTIVERSE PHASE PROJECTION OPERATOR

To describe observable universes from the complex continuum, we define the quantum phase projection operator P_{θ} :

$$P_{\theta} = \int_0^{2\pi} \delta(\theta - \arg(z)) d\theta \quad (\text{Eq. 9})$$

An individual observable universe U_{θ} corresponds to a specific phase angle θ in $[0, 2\pi)$:

$$g_{\mu\nu}^{\theta}(x) = \text{Re}(g_{\mu\nu}(x e^{i\theta})) \quad (\text{Eq. 10})$$

Thus, different cosmic phases belong to one unified geometry rather than disjoint regions of space.

V. COSMOLOGICAL IMPLICATIONS: DARK ENERGY & DARK MATTER

Evaluating the imaginary part of the metric tensor $\text{Im}(g_{\mu\nu})$ reveals an intrinsic vacuum stress-energy density:

$$\rho_{\text{vacuum}} = (c^4 / 8\pi G) \nabla^{\mu} \text{Im}(g_{\mu\nu}) = \rho_{\text{dark energy}} \quad (\text{Eq. 11})$$

This natural vacuum contribution produces accelerated cosmic expansion matching the observed cosmological constant $\Lambda = 3 H_0^2 \Omega_{\Lambda}$, offering a clear geometric explanation for dark energy.

VI. EMPIRICAL PREDICTIONS & EXPERIMENTAL VERIFICATION

1. Gravitational Wave Ringdown Phase Shifts: Next-generation detectors like LISA and Einstein Telescope can search for small imaginary phase perturbations $\delta\phi \sim 10^{-21}$ rad during binary black hole mergers.

2. Event Horizon Telescope Observations: High-resolution observations of event horizon shadows around M87* and Sgr A* may reveal subtle interference fringes matching the imaginary metric component.

VII. CONCLUSION

We have presented a clear, unified mathematical model demonstrating that the multiverse can be understood as phase projections of a single four-dimensional complex spacetime manifold. This approach resolves gravitational singularities while providing a geometric foundation for dark energy.

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